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Finally, realizing Synergies **costs money**. There will always be “integration costs” associated with a deal, and certain types of Synergies, such as headcount reductions, will cost even more due to severance costs.

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Merger Models – Calculations

You can determine whether a deal will be accretive or dilutive by comparing the Weighted Cost of Acquisition to the Seller’s Purchase Yield (Net Income / Purchase Equity Value), as defined in the previous section.

If you get a question about the Combined Equity Value, Combined Enterprise Value, or Combined Multiples, the rules are shown below.

Note that these rules **ignore** synergies and acquisition effects beyond the Interest Expense on New Debt, Foregone Interest on Cash, and New Shares Issued:

1. **Combined Equity Value** = Acquirer’s Equity Value + Market Value of Stock Issued in the Deal (but this may change post-announcement if the Acquirer’s share price falls).
2. **Combined Enterprise Value** = Acquirer’s Enterprise Value + Purchase Enterprise Value of Target (but this may change post-announcement if the Acquirer’s share price falls).
3. **Combined Valuation Multiples:** TEV-based Combined Multiples are in between the Acquirer’s current trading multiples and Target’s purchase multiples. Equity Value-based Combined Multiples are usually in this range as well, *but they do not have to be*. TEV-based Combined Multiples are not affected by the deal financing, but Equity Value-based Combined Multiples are.

These questions aren’t *that important*, but many groups like to ask them if you’ve had previous banking or M&A experience.

1. Company A has 10 shares outstanding at a share price of \$25.00, and its Net Income is \$10.

It acquires Company B for a Purchase Equity Value of \$150. Company B has a Net Income of \$10 as well. Both companies have 25% tax rates. How accretive is a 100% Stock deal?

Company A’s EPS is $\$10 / 10 = \1.00 .



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Company A must issue 6 new shares to do the deal since $\$150 / \$25.00 = 6$, so the Combined Share Count is $10 + 6 = 16$.

Since no Cash or Debt is used and the tax rates are the same, the Combined Net Income = Company A Net Income + Company B Net Income = $\$10 + \$10 = \$20$.

Therefore, the Combined EPS is $\$20 / 16 = \1.25 , so it's accretive by 25% (**mental math:** 25% of 16 is 4, and 20 is 4 greater than 16, so take the \$1.00 EPS and increase it by 25%).

2. Suppose that Company A now decides to use 100% Cash or 100% Debt to do this deal. If the Cash interest rate is 4%, and the Debt interest rate is 10%, will this deal be accretive or dilutive under these financing structures?

The After-Tax Cost of Cash is $4\% * (1 - 25\%) = 3\%$, and the After-Tax Cost of Debt is $10\% * (1 - 25\%) = 7.5\%$.

The Seller's Purchase Yield is $\$10 / \$150 = \sim 6.7\%$.

Therefore, a 100% Cash deal will be accretive, and a 100% Debt deal will be dilutive.

3. Company A has a P / E of 11x, a Debt Interest Rate of 8%, a Cash Interest Rate of 4%, and a Tax Rate of 25%.

It wants to acquire Company B at a purchase P / E multiple of 20x using 1/3 Stock, 1/3 Debt, and 1/3 Cash. Will the deal be accretive?

Company A's After-Tax Cost of Stock is $1 / 11$, or $\sim 9\%$, its After-Tax Cost of Debt is $8\% * (1 - 25\%) = 6\%$, and its After-Tax Cost of Cash is $4\% * (1 - 25\%) = 3\%$.

Company B's Purchase Yield is $1 / 20$, or 5% .

The Weighted Cost of Acquisition is $9\% * 1/3 + 6\% * 1/3 + 3\% * 1/3 = 3\% + 2\% + 1\% = 6\%$.

The deal will be **dilutive** since the Weighted Cost exceeds Company B's Purchase Yield.

4. Company A, with a current P / E multiple of 20x, acquires Company B for a P / E purchase multiple of 10x using 100% Debt.

What interest rate on Debt would make the deal dilutive? Assume a 25% tax rate.



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Company B's Yield is $1 / 10$, or 10%, so the After-Tax Cost of Debt must be *above* 10% for the deal to be dilutive.

At a 25% tax rate, $10\% / (1 - 25\%) = \sim 13.3\%$, which you can round to "Around 13%" (**mental math**: 75% of 10% is 7.5%, but we're trying to find the number that 10% is 75% of → So it must be higher by at least 2.5%, which you can round up to $\sim 13\%$).

Most companies do not pay 13% interest rates on their Debt, so a 100% Debt will almost certainly be accretive.

5. Company A has an Equity Value of \$2,000 and a Net Income of \$200. Company B has a Purchase Equity Value of \$1,200 and a Net Income of \$100.

How much in Synergies must be realized for a 100% Stock deal to be accretive?

Company A's P / E is $\$2,000 / \$200 = 10x$, so its Cost of Stock is $1 / 10 = 10\%$. Company B's Purchase P / E is $\$1,200 / \$100 = 12x$, so its Yield is $1 / 12$, or $\sim 8.3\%$.

Therefore, without Synergies, this deal would be slightly dilutive.

Company B's Purchase Yield must exceed 10% for the deal to turn accretive. This means its Purchase P / E multiple must be below 10x, which means its Net Income must be above \$120 (since $\$1,200 / \$120 = 10x$).

Therefore, there must be **\$20 in After-Tax Synergies** for this deal to be accretive. At a 25% tax rate, that means $\sim \$27$ in Pre-Tax Synergies.

6. Continuing with this same example, Company A is paying a 20% premium for Company B, so Company B's Current Equity Value is \$1,000.

Suppose the market doesn't "like" this 20% premium, and investors sell off Company A's stock.

How would Company A's stock price change? Assume it has 200 shares outstanding at \$10.00 per share and assume that Company B gets a fixed share count in the deal.

The Combined Equity Value in a deal normally equals the Acquirer's Equity Value + Stock Issued in the Deal, so it's $\$2,000 + \$1,200 = \$3,200$ here before the market's reaction.

If the market does *not* believe in this premium, the Combined Equity Value will fall by the premium paid, so it will go from \$3,200 to \$3,000 here.



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We know that Company A has 200 shares and that Company B gets $\$1,200 / \$10.00 = 120$ shares because the question states “fixed share count.”

Therefore, the post-market-reaction share price is $\$3,000 / 320 = \sim\9.38 , which you could round to “between \$9.00 and \$10.00” (**mental math:** $\$3,000 / 300 = \10.00 and $\$3,000 / \sim 333 = \9.00).

The **intuition** is that this drop of \$0.62 is $\sim 6.2\%$ of Company A’s share price, and \$200 is $\sim 6.2\%$ of the \$3,200 Combined Equity Value.

7. Continuing with the same scenario, your co-worker reviews the changes and claims that Company A's share price would not have fallen if this had been a 100% Cash or Debt deal.

Are they correct?

No. In this case of a “disappearing purchase premium,” both the Combined Equity Value and the Combined Enterprise Value fall by the purchase premium *regardless of the financing*.

In this example, Company B still gets a \$1,200 Purchase Equity Value in a 100% Cash deal, even if the market disapproves of this 20% premium.

So, Company A still pays \$1,200 of Cash to do the deal, which means the Combined Cash balance is lower by \$1,200. That normally boosts the Combined Enterprise Value by \$1,200.

But in this case, Company A’s Equity Value will fall by \$200 to reflect the “disappeared purchase premium,” so the Combined Enterprise Value will be up by \$1,000 rather than \$1,200.

The only difference is that if it’s a 100% Cash or Debt deal, Company A’s share price falls by \$1.00 rather than \$0.62 since $\$200 / 200 \text{ shares} = \1.00 .

8. An Acquirer with an Equity Value of \$500 million and an Enterprise Value of \$600 million buys another company for a Purchase Equity Value of \$100 million and a Purchase Enterprise Value of \$150 million.

What are the Combined Equity Value and Enterprise Value in a 100% Stock deal?

The Combined Enterprise Value equals the Enterprise Value of the Buyer plus the Purchase Enterprise Value of the Seller, so it’s $\$600 + \$150 = \$750$ million.

If it’s a 100% Stock deal, the Combined Equity Value = Acquirer’s Equity Value + Value of Stock Issued = $\$500 + \$100 = \$600$ million.



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9. How do these figures change if it's a 50% Cash / 50% Stock deal instead?

The Combined Enterprise Value stays the same regardless of the deal financing, so it's still \$750 million.

The Combined Equity Value changes based on the financing because it equals the Acquirer's Equity Value + Value of Stock Issued.

So, in a 50% Cash / 50% Stock deal, the Combined Equity Value is $\$500 + \$50 = \$550$ million.

10. An Acquirer with an Equity Value of \$500 million and an Enterprise Value of \$600 million has a Net Income of \$50 million and an EBITDA of \$100 million.

The Target, with a Purchase Equity Value of \$100 million and a Purchase Enterprise Value of \$150 million, has a Net Income of \$10 million and an EBITDA of \$15 million.

What are the Combined P / E and TEV / EBITDA multiples in a 100% Stock deal? Assume 25% tax rates for the Acquirer and Target.

The Combined Equity Value in a 100% Stock deal is $\$500 + \$100 = \$600$ million, and the Combined Enterprise Value is $\$600 + \$150 = \$750$ million.

The Combined EBITDA is \$115 million, and the Combined Net Income, assuming the same tax rates and no interest effects since it's a 100% Stock deal, is $\$50 + \$10 = \$60$ million.

Therefore, the Combined P / E multiple is $\$600 \text{ million} / \$60 \text{ million} = 10x$, and the Combined TEV / EBITDA multiple is $\$750 \text{ million} / \$115 \text{ million} = \sim 6.5x$.

11. How do the multiples change if this is a 100% Debt deal funded with an 8% interest rate issuance?

The Combined EBITDA and Enterprise Value stay the same because they do not depend on the deal financing, so the Combined TEV / EBITDA is still $\sim 6.5x$.

The Combined Equity Value is now just the Acquirer's Equity Value of \$500 million.

The Combined Net Income equals $\$50 + \$10 - \$100 * 8\% * (1 - 25\%) = \$50 + \$10 - \$6 = \$54$.

Therefore, the Combined P / E multiple is now $\$500 \text{ million} / \$54 \text{ million} = \sim 9.3x$, which you can estimate as "between 9x and 10x" (since $\$500 / \$50 = 10x$ and $\$500 / \sim \$56 = 9x$).